



Sequences of Polyhedra

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as a group of automorphisms on F^{d+1} and lifting the action of G on $P^d(F)$. By Theorem 2, there is a fixed point for T in F^{d+1} ; hence there is one for G in $P^d(F)$. We have proved:

THEOREM 3. *Let F be an algebraically closed field of characteristic p , G a group with p^n elements acting as F -automorphisms of $P^d(F)$. Then there is a point of $P^d(F)$ left fixed by G .*

5. A characterization of p -groups. A. R. Magid and I. Reiner have pointed out to us that there is a converse to Theorem 2.

THEOREM 4. *Let F be a field of characteristic $p \neq 0$, G a finite group. If every nonzero left FG -module has a fixed point, then G is a p -group.*

Proof. The number r of nonisomorphic irreducible FG -modules is the number of conjugacy classes of p' -elements of G , i.e., of elements of G whose order is relatively prime to p (see [8, p. 17, Theorem 1.5 and p. 31, Example]). If G is not a p -group, then $r > 1$. But F is the only irreducible FG -module with a fixed point.

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UNSOLVED PROBLEMS

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SEQUENCES OF POLYHEDRA

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In the Euclidean plane, let P_0 be a convex polygon, P_1 the polygon spanned by the side-midpoints of P_0 , P_2 the polygon spanned by the side-midpoints of P_1 , and so on. Cadwell

[1], [2] proved that in the sequence $P_0, P_1, P_2, \dots$ the polygons will be more and more similar to the affine image of a regular polygon. More precisely, there is a set of affine transformations $a_0, a_1, a_2, \dots$ such that the limiting figure of the set $a_0 P_0, a_1 P_1, a_2 P_2, \dots$ is a regular polygon.

This nice result remained almost unnoticed until it was partly rediscovered by this author [3], who gave a geometrical proof for polygons with less than seven sides and suggested several analogous problems. The paper [3] started a set of investigations dealing with various generalizations and variants of the original problem, giving new proofs and rediscovering old ones [4]–[13]. The sequence of polygons $P_0, P_1, \dots$ was studied, independently of Cadwell, also in the papers [14] and [15].

In 3-space, problems of another type present themselves. Given a convex polyhedron P we can consider, for instance, the convex hull of the centroids of the faces of P . Starting with an affinely regular solid, iteration of this process yields smaller and smaller copies of the solid and its dual. But a small distortion of the starting solid may cause a radical change in the sequence of polyhedra. Starting, for instance, with a polyhedron topologically isomorphic to the regular octahedron, in general we will obtain, in the first step, a polyhedron with 8 vertices and $2 \cdot 8 - 4 = 12$ triangular faces; in the second step, a polyhedron with 12 vertices and $2 \cdot 12 - 4 = 20$ triangular faces, and so on. Thus this process has not the effect of making the shape of the polyhedron more and more regular, only more and more complicated. Omitting the problems arising in connection with this process, we consider another analogue of the above sequence of polygons.

Let P_0 be a convex polyhedron, and let P_i ($i = 1, 2, \dots$) be the convex hull of the *edge*-midpoints of P_{i-1} . Since $P_i \subset P_{i-1}$, and P_i contains the centroids of the faces of all of its predecessors $P_{i-1}, \dots, P_0$, the limiting figure $\lim_{i \rightarrow \infty} P_i$ is a nondegenerate convex body which we call the **kernel** of P_0 . From among the various problems concerning the kernel of a convex polyhedron, we call attention to the following ones:

1. Prove or disprove the conjecture that among the convex polyhedra of equal volume the kernel of the tetrahedron has the least possible volume.
2. What can be said about the structure of the boundary of the kernel? Is the kernel of any convex polyhedron smooth?

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